

to bring in that type of intelligence onto the edge became relevant,” as per Venkateswar Melachervu, CTO, Brillium Technologies. He says, “Edge is where data resides and is collected.”

AI on edge is a system that has AI capabilities on the device itself or at some local node. This innovative feature allows devices to make their own decisions based on their previous learnings of different situations and their outcomes.

For instance, devices such as smartphones, smartwatches—even a temperature sensor—can be classified as edge devices. It is to be understood that edge computing is needed not just for the industry but also for the consumer.

“A lot of data is driven from machines or IoT devices, and that data needs to be computed from the security and privacy point of view. Therefore we need to have a better understanding of the differences between IoT, cloud, and edge computing,” opines Dharmendra Kumar, head of IoT, Arihant.

Since intelligence on edge devices such as smartphones, smartwatches, and connected cars has been there for a long time, Rishi Gargi, founder, Thingify adds another important point, “A boundary needs to be established between what is AI on edge and what other technologies are, and how AI on edge is leveraging AI without having any Internet or making decisions on its own.”

## Why go for AI on edge

The power of today’s computing is much more than what was available earlier. Even the latency and longevity of the batteries have improved. All these essentially have helped AI to be pushed for running on the edge, be it IoT devices, within the gateways, or the edge network.

Edge computing generally incorporates a processor and storage, as well as logic to process information. An example is Alexa, which uses natural language processing (NLP) at

the edge to respond to queries from a human being. But AI on edge is more than just computing. “It is not just about processing the data, but also deriving trends and patterns or intelligence by applying machine learning algorithms to give immediate feedback to the user,” thinks Vinay Solanki, head of IoT, Napino.

An example could be of a smart camera that has been installed in a smart city application for monitoring vandalism. While capturing videos or images, it could decide locally without sending any feedback to the cloud and thus become smarter on its own.

## Data security is the biggest concern

The real challenge for a customer is data security. People do not want to save their data on the cloud. When running an algorithm in the cloud on a stored data, it should be properly safeguarded, both on the device from where it has been generated and in the network. Cloud can enforce certain other security protocols like firewall, some encryption-decryption logic, or public-private key for the handshake, and make it safer.

“Some users might not be willing to share their data on the cloud. So, this approach can transform the capabilities of such applications to store data on mobile as well as process it. Health-related applications are extremely sensitive, therefore it is important to process the data on the mobile,” believes Amit Kurhekar, head of Data Science, Yodlee.

“It may sound a little oxymoronic because when you have distributed computing resources in the edge network, you are exposed to data attack (for instance DDoS). But usually, AI-edge is considered more secure for your data,” as per Venkateswar Melachervu.

Nikhil Bhaskaran believes that since people do not want all of their private data to go on the cloud, the edge is becoming popular.

## The cost concern

Having AI on cloud vs AI on edge is an important question, because on the edge it is the responsibility of the device and the device designer to ensure that it is secured. If somebody hacks into the device, it should be tough for the person to extract the data. But that might result in more cost of the device and more processing power requirement.

“You have to look at the ROI (return on investment) of the application that you are thinking of. One should not apply IoT just anywhere. It should make sense from a use-case perspective. Same goes for AI on edge. If you want to make the edge smarter, then consider whether it is an ROI-friendly decision, because if you need to invest to make every device smarter, then with that there will be added complexity of making the solution secure. One needs to be clear whether any extra investment is going to justify the output or the ROI that will be derived from the solution. It all depends upon what kind of real-time decision you need to make at the edge, or whether sending the feed to the cloud or network is going to be costly, time-consuming, or not secure,” advises Vinay Solanki.

Amit Kuhrekar advises having sufficient funds whenever you are developing something. Consider the cost of development, cost of hardware, and how many such devices are going to be needed. At the end of the day, you will have to maintain that.

“Always factor in the cost of your solution in the design process itself,” opines Nikhil Bhaskaran, founder, Shunya OS.

In terms of cost-effectiveness, it is a business decision whether to have an aggregated data model in a manufacturing unit. If you have twenty machines, then it is better to deploy sensors at each machine, which may be cost-effective (depending on the business need). The sensors could send data using machine-to-machine communication to a central node